Moving Objects Have Energy

Chapter 2

Everywhere you look, you see things move. Riders pedal bikes. People walk and run. Birds fly. Fish swim along the bottom of the ocean. Rain falls from the sky. Objects move in deep space too. Planets orbit stars. Meteoroids move through space. Another word for the movement of objects is **motion**. Motion occurs when something changes position.

Big Question

How are energy, change, and movement of objects related?

Vocabulary

motion, n. the process of an object changing position



The ball's position changes as it falls through the hoop.

Think About Describing Motion

Moving objects are described by how their position is changing. Scientists describe how fast an object is moving and changing position. Scientists also describe the direction in which an object is moving—left, right, up, down, and so on.

Moving Objects Can Cause Changes

Some objects, such as the basketball players in the picture, cause their own motion by moving their muscles. They jump and run from one end of the court to the other. The movement of muscles causes change.

Other objects, such as the basketball, are put into motion by outside motion. Moving basketball players cause the ball to change position. The player's hand dribbles the ball up and down. Players pass and catch the ball across the court. In all these examples, moving objects cause changes.



The players and the ball in a basketball game are in constant motion, and the motion causes things to change.

Moving Objects Have Energy

When a moving basketball hits a net, *swish*, the net moves. The net changes from sitting still to moving. When a skier slides down a slope of fresh snow, *whoosh*. The snow changes from powdery to hard and packed beneath the skis. When falling raindrops hit the surface of a smooth puddle, *splash*! The surface of the puddle changes from glassy and still to rippling with motion.

Flying basketballs, speeding skiers, falling raindrops, and all other moving objects have energy. How can you tell? You can tell because moving objects can cause change, and energy is the ability to cause a change. All moving objects have **energy of motion**.

Vocabulary

energy of motion, n. the energy an object possesses while it is moving



The motion energy of a falling raindrop changes the surface of a puddle.

People Use the Energy of Motion

Hammers and Nails

Because moving objects can cause a change, a moving object can be useful. People have figured out many ways to use the energy of objects in motion. We cut our food by slicing it with knives, and we cut our grass with spinning lawn mower blades. These objects in motion can cause the changes that result in getting something done.

Imagine trying to push a nail into a block of wood with your hands. That would be nearly impossible to do. Now imagine placing a hammer gently against the head of the nail and pushing the nail into the wood. You still would not move the nail into the wood. But swing the hammer against the head of the nail—*BANG*!



The motion energy of the hammer causes a change when the hammer hits the nail.

The hammer is heavy, and the swing is fast. The moving hammer now has energy of motion. That energy can change the position of the nail, even forcing it into the wood.

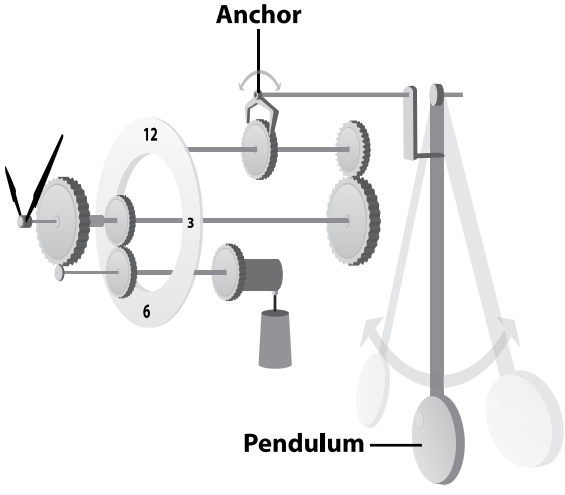
Energy from the moving hammer drives the nail into the wood, but what moves the hammer? The motion of the builder's arm provides the energy to change the position of the hammer.

Pendulum Clocks



Engineers, such as clockmakers, have applied the energy of motion for centuries. Clockmakers as early as the 1650s used their knowledge about the energy of a swinging pendulum to help tell time.

Look at the diagram of a clock shown here. The pendulum on the clock swings back and forth, back and forth, *tick-tock*. As the pendulum swings, it causes changes to the position of the clock's gears. The gears, as a result, rotate the hands of the clock. Those hands point to the numbers around the clock's face, allowing us to tell the time.



Think About Swinging Motion

The energy of motion of the pendulum arm causes changes in the gears, but what moves the pendulum arm? Gravity! Gravity pulls the arm down as it swings from its highest points on either side. A pendulum clock also has weights or springs that help pull or push on the pendulum.

Wind Turbines

Wind is moving air. Like everything that is in motion, wind has energy. Long ago, people discovered they could use the motion energy from wind to cause changes and get things done. People used wind to push the sails of ships to move across the sea.

Wind has also long been used to push the sails or blades of windmills. Early windmills captured the wind's motion energy to turn the blades. Then, the moving parts inside windmills could, as a result, grind grains or pump water.

Modern wind turbines work in much the same way. Wind turbines use the energy from moving air to spin giant blades. The rotating blades move internal parts that produce electricity, another form of energy.



Both traditional windmills and modern wind turbines use energy from moving air in helpful ways.